Auditing spatial information leakage in ecological prediction: quantifying the optimism in map accuracy with the `spLeakage` R package

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Abstract

1. Ecologists increasingly turn field observations into wall-to-wall maps—of species occurrence, abundance, biomass, traits and disease risk—and judge those maps by an accuracy estimated through cross-validation (CV). When observations are spatially autocorrelated, ordinary random CV places near-duplicate neighbours in both training and test sets, so the model is graded on near-interpolation rather than on the extrapolation the map is actually used for. Reported accuracy is then *optimistically biased.* Methods that *prevent* this by building spatially separated folds are well established, but no tool *audits* the split an analyst has already used (or a published result), or reports *how much* the accuracy is inflated.

2. The field is also divided: nearest-neighbour-matching CV treats random CV as optimistic for clustered data, whereas design-based critics show that for probability samples random CV is unbiased and spatial CV is *pessimistic.* Practitioners have had no way to tell which regime they are in.

3. We resolve both issues with one principle: spatial leakage and optimism are well-defined only *relative to a declared sampling design, estimand and prediction target.* The same split can be leaking, correct, or pessimistic according to those declarations—which dissolves the controversy (each side is correct in a different setting) and makes optimism computable.

4. On this basis we introduce a *signed* Spatial Leakage Index that audits any split and maps where it leaks; give optimism a formal estimand (excess explained variance under a Gaussian process) with a closed form and a Wasserstein bound; and provide a refit-free, simulation-calibrated estimator of accuracy inflation that flags when it is asked to extrapolate. All are implemented in the open-source `spLeakage` R package, which *audits rather than generates* folds and interoperates with existing spatial-CV tools.

5. Validated against exact Gaussian-process algebra, the index is a near-sufficient statistic for true optimism ($r = 0.98$) and recovers the *direction* of bias that unsigned matching statistics cannot ($r = 0.98$ vs -0.09). Auditing 16 real spatial datasets, reported accuracy is optimistic in **every** case (median 22%), and optimism splits into an autocorrelation component that spatial CV detects and a trend-extrapolation component it misses. `spLeakage` gives ecologists a practical instrument to detect, quantify, correct and pre-empt this overconfidence, complementing emerging reporting standards for predictive models.

Keywords: area of applicability; cross-validation; map accuracy; optimism bias; reproducibility; spatial autocorrelation; species distribution models; validation.

Plain language summary

Ecologists routinely turn survey data into maps and report how accurate those maps are. We show that the usual way of checking accuracy makes maps look better than they are, because nearby survey points end up in both the “learning” and the “testing” data. We provide a free R package, `spLeakage`, that takes a map-maker’s existing analysis and reports how inflated the accuracy is, where the problem is worst, and what to do about it—and we show that across many real datasets reported accuracy is too high every time, by about a fifth on average.

1 Introduction

Predictive maps are now a primary product of quantitative ecology. Species distribution models, abundance and biomass surfaces, habitat-suitability layers, trait maps and disease-risk surfaces are all built by training a statistical or machine-learning model on georeferenced observations and predicting across a landscape (Araújo et al., 2019; Zurell et al., 2020). The scientific and management value of such a map hinges on one number: its estimated predictive accuracy. If that number is wrong, decisions built on the map—reserve placement, risk targeting, change detection—are miscalibrated in ways that are invisible to the user.

For spatial data this number is systematically unreliable. Ecological observations are spatially autocorrelated: nearby sites resemble one another. Under an ordinary random train/test split, most test points have a training neighbour only a short distance away, so the model is evaluated on a task close to *interpolation*. At deployment, however, the map must predict at locations far from any observation. The reported error therefore understates the true error—it is *optimistically biased*—an instance of information *leakage* from training to test (Roberts et al., 2017; Ploton et al., 2020; Kattenborn et al., 2022). Ploton et al. (2020) showed that large-scale forest biomass maps are far less accurate than reported once spatial structure is respected; Kattenborn et al. (2022) documented the same inflation for convolutional networks on vegetation imagery. Overconfidence in map accuracy is, in short, a recognised and consequential problem in ecology.

What exists, and what is missing. A mature toolkit *prevents* leakage by constructing spatially or environmentally separated folds: spatial block CV (Roberts et al., 2017; Valavi et al., 2019), buffered / spatial leave-one-out CV (Le Rest et al., 2014; Telford and Birks, 2009) and nearest-neighbour distance matching CV (NNDM/kNNDM; Milà et al., 2022; Linnenbrink et al., 2024), in packages such as `CAST` (Meyer et al., 2024), `blockCV` (Valavi et al., 2019) and `spatialsample`. These are *fold generators*: they answer "how should I cross-validate?". None answers the question an analyst who has already run a model—or a reader assessing a published map— actually asks: *how much does my split leak, and how inflated is the accuracy I am about to report?* Reporting standards for distribution models now ask authors to state how they validated (Araújo et al., 2019; Zurell et al., 2020), but there has been no instrument to *audit* that validation and quantify its bias.

A live controversy. There is also genuine disagreement about what "correct" validation is. Milà et al. (2022) and Linnenbrink et al. (2024) argue the CV geometry should imitate the prediction geometry, making random CV optimistic for clustered data. Wadoux et al. (2021) argue, from a design-based standpoint, that for a *probability sample* random validation is unbiased and spatial CV is *pessimistically* biased (see also de Bruin et al., 2022). **Both are right, in different settings**—yet practitioners have no way to identify their setting, and so no principled basis for choosing.

This paper. We argue the gap and the controversy share a cause and a cure. Spatial leakage and optimism are not properties of a dataset or a split alone; they are defined only relative to a declared sampling *design*, *estimand* and prediction *target* (Section 2). Making these explicit dissolves the controversy and renders optimism computable. We then contribute: (i) a *signed* Spatial Leakage Index that audits any split and maps where it leaks (Section 3); (ii) a formal estimand and theory for optimism, with a closed form and a distribution-shift bound, presented for the general reader in Box 1; (iii) a refit-free estimator of accuracy inflation that flags its own extrapolation (Section 4); and (iv) design-aware guidance, bias-corrected accuracy, and optimal validation-survey design (Section 5). Everything is implemented in the open-source `spLeakage` package (Section 6). Section 7 validates the approach with exact algebra, a benchmark against NNDM, real-data tests, a 16-dataset audit of field-wide overconfidence, and two real case studies—a species distribution model and a disease-prevalence map—that show the audit attributing leakage to different causes. Section 8 gives practical guidance (Box 2) and discusses scope and limitations.

2 A framework: leakage relative to design, estimand and target

Our organising principle is:

> *"Spatial leakage" and "optimism" are not properties of a dataset or a split in isolation. They are well-defined only relative to a declared sampling **design**, a declared **estimand**, and a declared prediction **target**. The same split can be leaking, correct, or pessimistic depending on these three declarations.*

The three declarations do distinct work.

Design. Whether the data form a probability sample, a designed cluster sample, or a convenience sample decides whether random validation is unbiased (Wadoux et al., 2021) or optimistic (Milà et al., 2022). Crucially, design-basedness is a fact about *how data were collected* and cannot be read off the coordinates: a designed cluster sample and a convenience sample can produce identical point patterns. Clustering statistics (Clark–Evans, Ripley's K) can therefore only *flag risk*; the design must be *declared*.

Estimand. *Mean map accuracy over a region* (a design-based quantity) and *conditional predictive skill at a location* (a model-based quantity) are different questions; we never compare validation schemes across them.

Target. Predicting within the sampled footprint, mapping wall-to-wall, or extrapolating to a new region imply different deployment geometries, and hence different "correct" validation geometries.

This makes optimism well-posed: it is the gap between reported accuracy and the accuracy under *the validation that is correct for the declared design* $\times$ *estimand* $\times$ *target*—not relative to any one method. For a probability sample with a population-mean estimand, the correct baseline is random validation, so a spatially blocked scheme registers as *pessimism* (negative optimism), exactly the design-based critique, now reported as such. Because the declarations are inputs rather than inferences, the circularity that would otherwise make optimism ill-posed disappears. This reconciliation is the conceptual core of the paper; the index, the estimator and the recommender are its instruments.

3 The signed Spatial Leakage Index

For a split, let $g_{\text{obs}}(t)$ be the distance from each test point t to its nearest training point during CV, and $f_{\text{pred}}(p)$ the distance from each deployment location p to its nearest observation; distances are geodesic for geographic coordinates and projected otherwise. Their empirical distribution functions are $\widehat{G}_{\text{obs}}$ and $\widehat{G}_{\text{pred}}$. Leakage means test points sit *closer* to training data during CV than deployment points will sit to the sample, i.e. $\widehat{G}_{\text{obs}}$ is shifted to smaller distances.

Headline (dependence) form. Because geographic distance is not the same as statistical dependence, we convert distances to the fitted spatial correlation $\rho(h)$ from the empirical variogram and define the mean *retained spatial information* under CV and at deployment,

$$c_{\text{obs}} = \tfrac{1}{n}\textstyle\sum_t \rho(g_{\text{obs}}(t))\,, \qquad c_{\text{pred}} = \tfrac{1}{m}\textstyle\sum_p \rho(f_{\text{pred}}(p))\,,$$

$$\boxed{\text{SLI}_\rho = c_{\text{obs}} - c_{\text{pred}} \in [-1, 1].} \tag{126}$$

SLI_ρ is bounded, dimensionless and *signed*: positive means CV retains more spatial information than deployment has (optimistic leakage), negative means pessimism, near zero means well matched. Under anisotropy or non-stationarity a directional or local ρ is used, so the index tracks dependence rather than raw geometry. Box 1 shows that SLI_ρ is the leading term of true optimism, which is what validates it.

Distance form and the bridge to NNDM. A distance-scale companion SLI_d is the signed area between the two distribution functions, normalised by the variogram range. Its unsigned magnitude equals the NNDM/kNNDM Wasserstein statistic W when the curves do not cross, making the relationship to existing methods explicit; a directionality ratio $\delta \in [-1, 1]$ warns when the curves cross and a single number should not be trusted.

Mapping the leak. The per-observation contribution $\ell(t) = \rho(g_{\mathrm{obs}}(t)) - c_{\mathrm{pred}}$ sums to SLI_ρ; mapping $\ell(t)$ shows *where* a split leaks and averaging within folds shows *which* folds leak. This decomposition of an arbitrary split—not just a single summary number—is the substantive advance over a fold generator. Variogram parameters are estimated, often from the very clustered data that leaks, so an optional Monte-Carlo routine propagates that uncertainty into a confidence interval on SLI_ρ.

Box 1. What optimism *is*, and why a geometric index predicts it

Model the response as a Gaussian process plus noise: $Y(s) = \mu + Z(s) + \varepsilon(s)$, with $\mathrm{Cov}(Z(s), Z(s')) = \sigma^2\rho(\|s - s'\|)$ and $\varepsilon \sim N(0, \tau^2)$. Write the total variance $V = \sigma^2 + \tau^2$ and the *signal proportion* $w = \sigma^2/V$. For optimal (kriging) prediction, the error at a location is V minus an *explained variance* term EV that grows the closer (more correlated) the location is to its training data. Averaging over the test set (predicted from CV training folds) and over the deployment target (predicted from the full sample), the constant V cancels and

$$\text{optimism} = \underbrace{\tfrac{1}{n}\textstyle\sum_t \mathrm{EV}(t \mid \text{CV training})}_{\text{CV explains}} - \underbrace{\tfrac{1}{m}\textstyle\sum_p \mathrm{EV}(p \mid \text{full sample})}_{\text{deployment explains}}. \tag{B1.1}$$

Optimism is exactly the excess explained variance CV enjoys over deployment (B1.1 holds for any such process; it is positive when CV is “easier” than deployment and negative—pessimism—when harder). Keeping each point’s nearest neighbour, at distance d, the pointwise loss is $L(d) \approx V[1 - w^2\rho(d)^2]$, and (B1.1) becomes

$$\text{optimism} \approx V\, w^2 \Big(\mathbb{E}_t[\rho(g)^2] - \mathbb{E}_p[\rho(f)^2]\Big), \tag{B1.2}$$

i.e. (up to the scale Vw^2) a difference of mean retained correlation between CV and deployment—*the Spatial Leakage Index.* The theory therefore predicts which quantities govern optimism (the correlation range and smoothness, the signal proportion w, and the two nearest-neighbour-distance distributions) and explains why a cheap geometric index can stand in for an expensive refit. Finally, because L is smooth and bounded, optimism is bounded by a Wasserstein-1 distance between the CV and deployment distance distributions, $|\text{optimism}| \leq \|L'\| \, W_1$, connecting spatial leakage to distribution-shift theory and giving SLI_d a rigorous interpretation as a bound. *Scope:* the single-neighbour form underestimates magnitude for dense data (corrected by a $\approx 1.6\times$ multi-neighbour factor); non-Gaussian responses and non-kriging learners are handled empirically by the estimator below, for which this theory gives the envelope.

4 From leakage to a number: estimating optimism

A leakage score is abstract; ecologists want the consequence in their own currency: *"my reported accuracy is $X\%$ too high"*. `spLeakage` provides two routes.

Refit route. Re-evaluate the same model under the design-matched correct scheme (e.g. spatial block CV for clustered data) and report the gap, using a metric appropriate to the response—RMSE/MAE for continuous responses, and proper scoring rules (Brier, log-loss, Poisson deviance) for occurrence, prevalence and count data, so the optimism is meaningful for the binary and count responses common in ecology. This is accurate but model-conditional and requires refitting.

Refit-free estimator. A predictor maps the data and split geometry (range, smoothness, signal proportion, sample size, clustering, the signed SLI, the declared design and target) to expected optimism. It is calibrated once on a large simulation study—shipped with the package—spanning Matérn fields, random / clustered / preferential sampling, Gaussian / Poisson / Binomial responses, and several learners and targets, and returns an estimate with a calibrated (conformal) interval in milliseconds, without refitting. Critically, the estimator carries its *own* area-of-applicability check (Meyer and Pebesma, 2021): when a dataset falls outside the envelope it was trained on, it *abstains* rather than extrapolate—a tool that polices over-extrapolation must not commit it. When it abstains, the refit route always applies.

5 Recommendation, correction and validation design

Design-aware recommendation. The framework becomes a decision aid that elicits the estimand, then recommends over design $\times$ target (Table 1). Three rules keep it honest: the design is *declared, not inferred from geometry*; every recommendation is backed by a citation or a simulation result; and guidance is conditional ("*if* your estimand is X under framework Y,

Table 1: Design × target recommendation (after the estimand is declared). The design is an elicited input; geometry only flags risk. Cells follow the cited literature and our simulations (Section 7).

Design \ target	**Interpolation**	**Wall-to-wall map**	**New-region extrapolation**
Probability / design-based	Random CV	Design-based estimator	Caution; AOA
Clustered / preferential	NNDM / buffered	NNDM / kNNDM	Block + AOA
Convenience / opportunistic	Buffered LOO	kNNDM	Block + AOA, flag risk

then…”), surfacing the assumption rather than quietly taking a side. It states plainly *when spatial CV is not appropriate.* A data-driven companion builds each candidate scheme on the user’s own data and ranks them by how closely each matches the declared deployment geometry.

Corrected accuracy. A de-leaked estimator reports the accuracy an analyst *should* publish, with a bootstrap interval, and can translate a single *published* number into its corrected counterpart without the original model— making the package usable for third-party review of existing maps.

Designing validation surveys. Inverting the leakage logic, the package allocates a budget of independent validation points to estimate true map accuracy with minimum variance (distance-stratified Neyman allocation with inclusion weights), turning leakage diagnosis into survey design.

6 The `spLeakage` package

`spLeakage` is an R ($\geq$ 4.1) package built on `sf` (Pebesma, 2018), with heavier adaptors (`rsample`, `mlr3`, `caret`, `blockCV`, `terra`, `gstat`) optional and guarded at run time. It *audits rather than generates* folds, interoperating with the spatial-CV ecosystem. The core verbs are `detect_leakage()` (the signed SLI), `estimate_optimism()` / `predict_optimism()` (the two optimism routes), `recommend_validation()` / `rank_cv_schemes()` (guidance), `deleak_estimate()` (correction) and `design_validation()` (survey design), with channel-specific detectors for grouped/duplicated-location, covariate-space, temporal, spatiotemporal and large-scale-trend leakage, each paired with a fix. `audit_workflow()` inspects an existing resampling object and `report_leakage()` produces a submission-ready scorecard. The package passes `R CMD check` cleanly, ships three tutorials, and has a documentation website.

Table 2: Exact Gaussian-process validation of the theory in Box 1 (192 configurations).

Claim	Result
Closed form (B1.2) predicts exact optimism	$r = 0.975$
SLI_ρ is a near-sufficient statistic for optimism	$r = 0.976$
Wasserstein bound holds (single-neighbour optimism)	100%
Multi-neighbour amplification (exact / single-neighbour)	median 1.56

```
library(spLeakage)
tgt <- prediction_target(grid = pred_grid, type = "grid")   # where the map is used
lk  <- detect_leakage(occ, split = my_folds, target = tgt,  # signed leakage index
                      response = "presence", n_boot = 300)
lk; plot(lk)                                                  # score, CI, ECDFs, map
predict_optimism(lk)                                          # "% accuracy inflated"
recommend_validation(occ, estimand = "prediction",          # what to do instead
                     design = "clustered", target = "grid")
```

Listing 1: Auditing an existing species-distribution-model validation in five lines.

7 Worked examples and validation

We validate each claim in turn; scripts to reproduce every result ship with the package.

7.1 The index is a near-sufficient statistic for optimism

Across 192 Gaussian-process configurations, with explained variance computed exactly from covariance algebra (Box 1), the unsigned closed form (B1.2) tracks true optimism at $r = 0.975$, and the package's SLI_ρ at $r = 0.976$—explaining $\approx 95\%$ of the variance in true optimism (Table 2; Fig. 2). The Wasserstein bound holds in 100% of configurations for the quantity it governs.

7.2 Signed leakage recovers what matching statistics cannot

The natural question—"is this just NNDM's W?"—is answered against exact optimism over 288 configurations, 64% of which are *pessimistic* (the design-based regime). The signed SLI_ρ predicts *signed* optimism at $r = 0.977$; the unsigned W at $r = -0.090$ (Table 3). W cannot sign the bias by construction, so a large W leaves an analyst unsure whether to trust their accuracy more or less; the signed index recovers the direction in 97% of cases and beats W on magnitude because it carries the signal proportion that scales optimism. NNDM solves a fold-construction problem;

Table 3: Predicting exact optimism from the signed SLI versus NNDM's unsigned W (288 configurations).

Task	signed SLI_ρ	NNDM W (unsigned)
Predict *signed* optimism	$r = +0.977$	$r = -0.090$
Predict magnitude	$r = 0.955$	$r = 0.672$
Direction (sign) agreement	97%	silent by construction

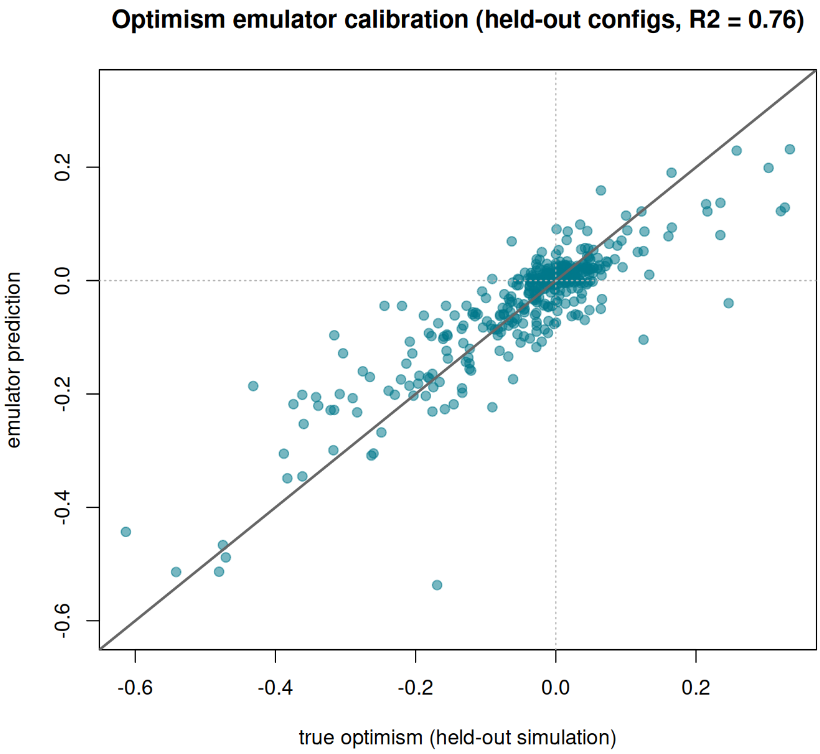

Figure 1: Out-of-sample calibration of the refit-free optimism estimator (held out by configuration): predicted versus true optimism with conformal intervals; $R^2 \approx 0.76$, interval coverage $\approx 94\%$, 97% of queries within the area of applicability.

`spLeakage` solves a diagnosis problem.

7.3 The estimator generalises, and reconciles the controversy

On a study of 1000 configurations held out by configuration, the refit-free estimator attains $R^2 \approx 0.76$, places 97% of queries inside its area of applicability, and its 90% intervals cover at $\approx 94\%$ (Fig. 1). The design effect that reconciles the debate reproduces in the data: clustered and preferential sampling shift random CV toward optimism (the Milà regime), while for well-covered targets with probability-like samples random CV is unbiased or slightly pessimistic (the design-based regime); the ordering is robust (Fig. 3).

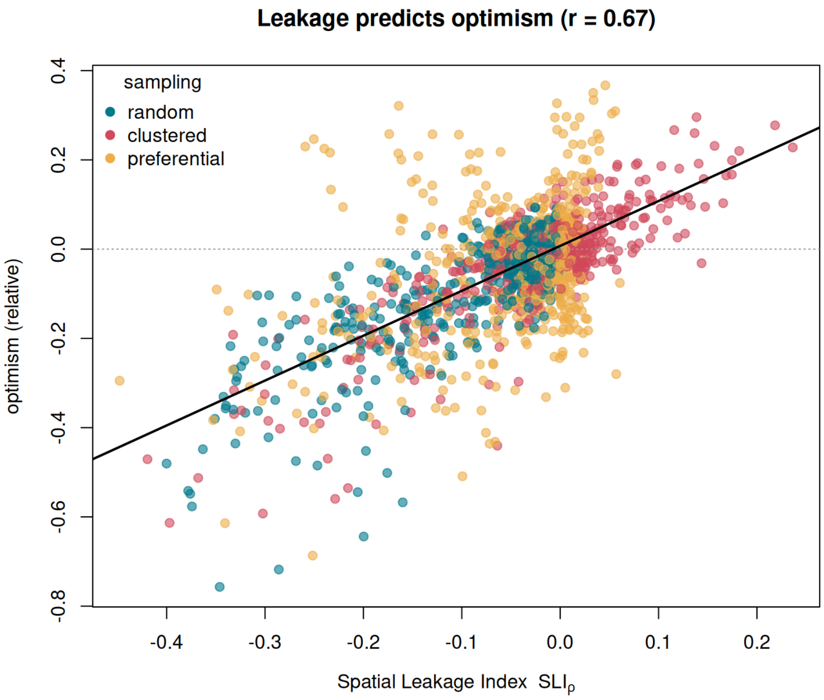

Figure 2: The Spatial Leakage Index tracks true optimism across the simulation grid (empirical cor = 0.67; $r = 0.98$ against exact optimism), the empirical counterpart of Box 1, eq. (B1.2).

7.4 Real fields: the problem is real and large

On three real spatial fields evaluated against genuinely independent held-out data (240 replicates), naive random CV understates true error by a median of 36% under extrapolation. The target-aware SLI separates extrapolation from interpolation deployments as real optimism does (cor = 0.52), and the de-leaked estimate recovers up to 89% of the true independent error—without ever seeing the held-out data.

7.5 How overconfident is ecological mapping? A 16-dataset audit

Using `spLeakage` as the instrument on 16 real spatial response variables (soil chemistry, heavy metals, rainfall, elevation, ozone and disease prevalence), with real optimism measured by spatial region-holdout, reported accuracy is optimistic in **every** dataset, with a median of 22% (interquartile range 17–34%) and more than 20% in two-thirds (Fig. 4).

The audit also reveals that optimism has *two* components. The autocorrelation component is what the SLI and NNDM detect. But the smooth, trend-dominated fields (rainfall, elevation) show the *largest* optimism (40–54%) with almost no short-range leakage signal: their error is the model's inability to extrapolate a large-scale *trend*, to which autocorrelation-based diagnostics

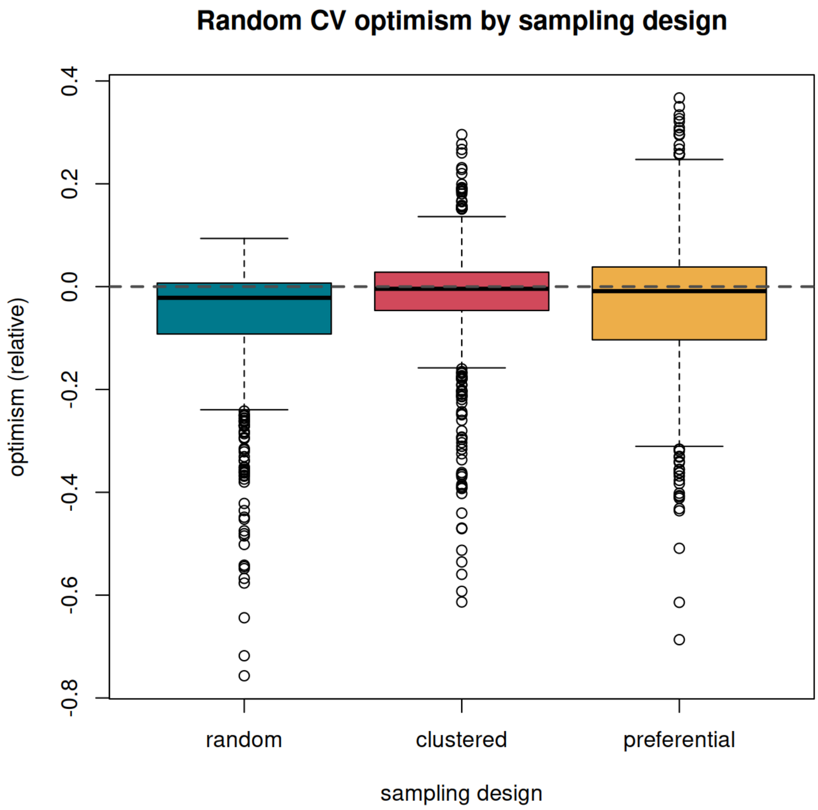

Figure 3: Reconciling the controversy. The optimism of random CV depends on sampling design and target: clustered/preferential designs push random CV toward optimism; probability-like samples on well-covered targets make it unbiased or slightly pessimistic. Both are cells of one framework, not contradictory claims.

are blind. A trend-strength feature recovers exactly this missed signal ($r = +0.75$). Field-wide overconfidence is therefore a two-channel phenomenon, and the 22% median is a *lower bound* on what autocorrelation-aware tooling alone would catch.

7.6 Case studies: two real maps, two leakage signatures

We close with two real applications that a single audit must be able to tell apart: a species distribution model whose optimism is genuine spatial autocorrelation, and a disease-prevalence map whose apparent leakage is in fact a survey-design artefact. Together they show `spLeakage` attributing leakage to the *right* cause.

Lead case: a species distribution model. We audit a presence/absence SDM built on the south-east Australia dataset of Valavi et al. (2019) (500 records; 243 presence; four bioclim covariates), predicting a wall-to-wall habitat-suitability map. An ecologist's default—a random 10-fold split validated with the Brier score—returns an optimistic split: $\mathrm{SLI}_\rho = +0.10$ (90% CI

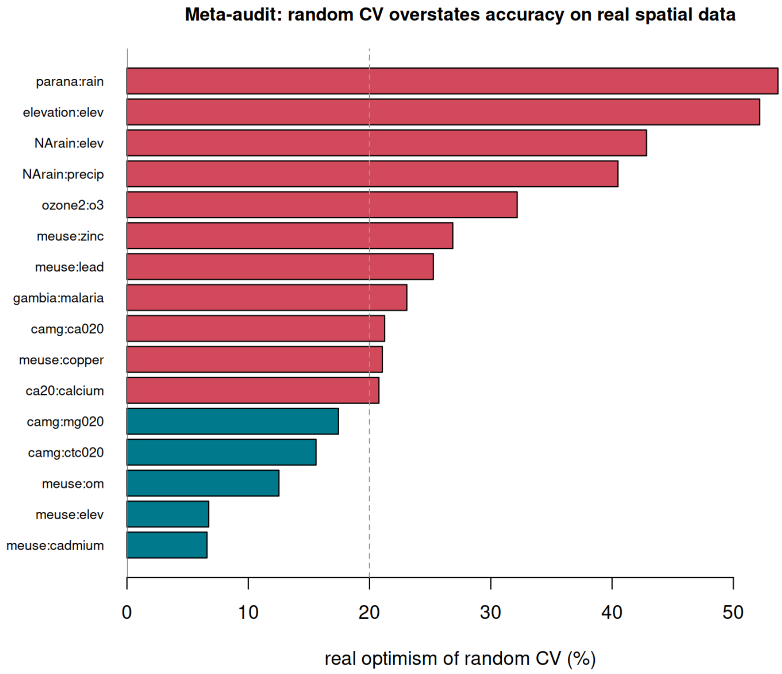

Figure 4: Auditing 16 real spatial datasets. Naive random CV is optimistic for every one (median 22%). The trend-dominated fields show the largest optimism yet the weakest short-range leakage—the second, trend-extrapolation channel.

$+0.07, +0.14$), and the Brier score is 42% **too good** relative to the design-matched block scheme. Two further channels sharpen the diagnosis. The covariate-space audit shows the map *extrapolates* the bioclim predictors well beyond where the model was tested (deployment covariate-space reach 3.73 vs test reach 1.04; signed feature leakage +2.7)—a concrete area-of-applicability warning for the suitability surface. Meanwhile the grouped-location channel finds *no* co-located duplicates, so this optimism is genuine autocorrelation, not a sampling artefact. Ranking candidate schemes on the data reproduces the paper's central nuance: random CV is optimistic (+0.10), buffered CV is *over-corrected into pessimism* (-0.22), and block CV best matches deployment (-0.004)—a distinction no static decision table provides.

Second case: mapping malaria prevalence. On geolocated *Plasmodium falciparum* prevalence surveys in Nigeria (Weiss et al., 2019) ($n = 66$), a naive random split gives $\mathrm{SLI}_\rho = +0.212$ despite a near-zero variogram signal—an apparent paradox the package resolves: the grouped-location channel attributes 21.2% of test points to leakage through *co-located repeat surveys*, deduplication collapses SLI_ρ to +0.006, and the grouped fix removes it entirely. The refit-free estimator correctly *abstained* (n below its training envelope), demonstrating the area-of-applicability guard on real data. Where the SDM's leakage was real autocorrelation, here it is a survey-design artefact—routine in ecological and epidemiological data with revisited sites (fixed plots, transect re-surveys, sentinel stations). The same multi-channel audit reaches opposite, correct verdicts in the two cases.

8 Discussion

A settled controversy and a practical workflow. The disagreement over spatial CV is not a conflict of facts but of settings. Once design, estimand and target are declared, optimism is a well-posed, signed quantity, and the data decide which prescription applies. `spLeakage` reports the setting and prescribes accordingly, conditionally rather than dogmatically (Box 2). It complements rather than competes with fold generators (`CAST`, `blockCV`, `spatialsample`) and with reporting standards (Araújo et al., 2019; Zurell et al., 2020): where those ask authors to *describe* their validation, this lets authors and reviewers *audit* it.

Scope and limitations. The theory in Box 1 is a Gaussian backbone: the single-neighbour form underestimates magnitude for dense data (correctable by a multi-neighbour factor), and non-Gaussian responses and non-kriging learners are handled empirically by the estimator, for which the theory gives the envelope. The estimator is honest about its own extrapolation and abstains when out of range. The most important open finding is the trend-extrapolation channel that all autocorrelation-based diagnostics miss; it is now flagged by a trend-strength feature but deserves a fully trend- and covariate-aware optimism model. Finally, design-basedness must be elicited: no geometric test can certify a probability sample, so the recommender is decision *support*, not automation.

Outlook. The distribution-shift view (Box 1) points toward model-agnostic optimism bounds beyond the Gaussian case, and coupling the autocorrelation and trend channels into one estimator is a clear next step. We hope the package becomes a routine step between fitting an ecological map and reporting its accuracy.

> **Box 2. A checklist for auditing your own map validation**
>
> 1. **Declare the deployment target.** Are you interpolating within the sampled area, mapping wall-to-wall, or extrapolating to a new region? (`prediction_target()`.)
> 2. **Declare the sampling design and estimand.** Is it a probability sample or opportunistic data? Do you want mean map accuracy or skill at a location? These are facts you know and the

coordinates do not.

3. **Audit the split.** Run `detect_leakage()`; read the *signed* index and its interval, and `plot()` where it leaks.
4. **Check other channels.** Revisited sites, covariate-space extrapolation, time, and large-scale trend (`audit_workflow()`).
5. **Quantify and correct.** Get the % inflation (`predict_optimism()` / `estimate_optimism()`) and report the de-leaked accuracy (`deleak_estimate()`).
6. **Act on the recommendation.** Use `recommend_validation()` / `rank_cv_schemes()`—and heed it when it says spatial CV is *not* appropriate.

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Conflict of interest

The author declares no conflict of interest.

Author contributions

OJ conceived the study, developed the methods and software, performed the analyses and wrote the manuscript.

Data availability and reproducibility

The `spLeakage` package and all analysis scripts that reproduce the results are openly available at `https://github.com/olatunjijohnson/spLeakage` (MIT licence), with documentation at `https://olatunjijohnson.github.io/spLeakage/`. All datasets are public (the south-east Australia presence/absence SDM data and bioclim covariates via `blockCV`; Malaria Atlas Project via `malariaAtlas`; `meuse`, `ca20`, Paraná and the remaining audit fields via `sp`, `geoR` and `gstat`). Code archival on Zenodo with a DOI will accompany acceptance.

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